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Liquid discharging apparatus

Abstract

A liquid discharging apparatus, having a main housing, a liquid discharging head with nozzles, a container to store a printable medium, a waste liquid storage located at a position to at least partly overlap the container and lower than the container in a vertical direction, a mountable compartment, in which the waste liquid storage is demountably mountable, is provided. The mountable compartment is formed on a sideward face of the main housing on one side in a widthwise direction being a scanning direction of the liquid discharging head. The mountable compartment is open at a position frontward with respect to a connector, which connects a power cord for receiving external power supply to the main housing there-through, in a front-rear direction being a conveying direction to convey the printable medium and toward which the printable medium is ejected from the main housing.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority from Japanese Patent Application No. 2022-105652 filed on Jun. 30, 2022. The entire content of the priority application is incorporated herein by reference.
BACKGROUND ART
(2) A liquid discharging apparatus that may discharge liquid from a liquid discharging head to print an image on a sheet is known. The liquid discharging apparatus may have a waste liquid container to collect waste liquid discharged from the liquid discharging head. The waste liquid container may be located below a feeder cassette and may be drawn outward for being exchanged in a same direction as the feeder cassette to be drawn outward from a body of the liquid discharging apparatus.

Description

DESCRIPTION
(1) In the liquid discharging apparatus, the direction to draw the feeder cassette outward may be the

same as a direction, in which the sheet is ejected outward from the body of the liquid discharging apparatus. In other words, the direction to draw the waste liquid container outward may be the same as the direction, in which the sheet is ejected outward from the body of the liquid discharging apparatus. Therefore, when the sheets are left in stack on an ejection tray, the waste liquid container drawn outward may interfere with the stacked sheets and may cause troubles in a user's works to exchange the waste liquid container with a new waste liquid container.
(2) The present disclosure is advantageous in that a liquid discharging apparatus, in which waste liquid containers may be exchanged easily while sheets stay in an ejection tray, is provided.
(3) FIG. **1** is a perspective exterior view of an inkjet printer.
(4) FIG. **2** is a cross-sectional view of an inner structure of the inkjet printer.
(5) FIG. **3** is a cross-sectional view of the inkjet printer viewed from an upper side.
(6) FIG. **4** illustrates an arrangement of a waste liquid box viewed from an upper side.
(7) FIG. **5** illustrates another arrangement of the waste liquid box viewed from the upper side.
(8) FIG. **6** illustrates the arrangement of the waste liquid box viewed from a front side.
(9) FIG. **7** illustrates another arrangement of the waste liquid box viewed from the upper side.
(10) FIG. **8** illustrates another arrangement of the waste liquid box viewed from the upper side.
(11) FIG. **9** is an illustrative view of an inkjet printer viewed from the front side.
(12) FIG. **10** is an illustrative view of the inkjet printer viewed from the upper side.
(13) FIGS. **11**A-**11**B are illustrative views of an intermediate box, the waste liquid box, and a waste liquid guide in the inkjet printer.
FIRST EMBODIMENT
(14) A first embodiment of the present disclosure will be described below.
(15) [Inkjet Printer **1**]
(16) <Overall Configuration>
(17) An overall configuration of an inkjet printer **1** will be described below with reference to FIGS. **1**-**3**. FIG. **1** is a perspective exterior view of the inkjet printer **1**. FIG. **2** is a cross-sectional view of an inner structure of the inkjet printer **1**. FIG. **3** is a cross-sectional view of the inkjet printer **1** viewed from an upper side.
(18) The inkjet printer **1** shown in FIG. **1** may discharge liquid. In the following description, directions in the inkjet printer **1** are defined with reference to an orientation of the inkjet printer **1** in a usable condition set on a base plane as shown in FIG. **1**: a vertical direction as indicated by upward and downward arrows in FIG. **1** includes up-to-down and down-to-up directions, whereas a side toward the base plane is defined as a lower side. A front-rear direction as indicated by arrows pointing lower-rightward and upper-rearward in FIG. **1** includes front-to-rear and rear-to-front directions, whereas a side on which an opening **20** is located is defined as a front side. A left-right direction, or a widthwise direction, as indicated by arrows pointing lower-leftward and upper-rightward in FIG. **1** includes leftward and rightward directions, whereas a lefthand side to a user who faces a face on the front side of the inkjet printer **1** is defined as a leftward side.
(19) In the following paragraphs, the vertical direction may be called a direction intersecting orthogonally with a nozzle surface **421** of a recording head **41** (see FIG. **3**). The front-rear direction may be called a conveying direction to convey a sheet P by a conveyer **5** (see FIG. **2**). The widthwise direction may also be called a scanning direction, in which a carriage **40** (see FIG. **2**) is movable. The recording head **41**, the conveyer **5**, and the carriage **40** will be described further below.
(20) As shown in FIG. **1**, the inkjet printer **1** includes a main housing **11** and a scanner housing **12** stacked on an upper side in the inkjet printer **1**. The main housing **11** and the scanner housing **12** together form a substantially rectangular-box shape.
(21) On a front side of the main housing **11**, an operation panel **13** and a cartridge cover **14** are arranged. The operation panel **13** includes operation devices, such as operation buttons, and a liquid crystal display. The cartridge cover **14** is pivotable with respect to the main housing **11**.

Inside the cartridge cover **14**, as shown in FIG. **3**, ink cartridges **141** are arranged and attached to a cartridge case **140**. Moreover, as shown in FIG. **1**, on the front side of the main housing **11**, the opening **20** is formed. Through the opening **20** formed on the front side of the main housing **11**, a feeder tray **21** for containing a printable medium and an ejection tray **22** may be attached to or detached from the inkjet printer **1**. The scanner housing **12** accommodates a scanner (not shown), which may read an image appearing on a sheet P.
(22) A waste liquid box exchangeable unit **15**, a power unit **16**, and a gripper **17** are located on a sideward face on a widthwise end of the main housing **11**. For example, the waste liquid box exchangeable unit **15**, the power unit **16**, and the gripper **17** may be located on a leftward face of the main housing **11**.
(23) The waste liquid box exchangeable unit **15** includes a cover **151**, which may be pivotable with respect to the main housing **11**, and a mountable compartment **152** (see FIG. **3**), which is exposed when the cover **151** is open and may accommodate a waste liquid box **60**. In other words, the cover **151** being open allows the waste liquid box **60** to be mounted in and demounted from the main housing **11**. In the following paragraphs, directions, in which the waste liquid box **60** is mountable in and demountable from the main housing **11** may be called a mountable/demountable direction.
(24) The mountable compartment **152** is located at a position such that the waste liquid box **60** is mounted at a position at least partly lower than the feeder tray **21** and the ejection tray **22** and overlap the feeder tray **21** and the ejection tray. In other words, the mountable compartment **152** is at least partly located in the vertical direction to be lower than the feeder tray **21** and the ejection tray **22**. The mountable compartment **152** has a form to fit with the waste liquid box **60** and has a wall, on which a sideward face of the waste liquid box **60** may abut, on an inner side in the mounting/demounting direction. The wall may therefore restrict the waste liquid box **60** from moving further inward. Optionally, the mountable compartment **152** may have a guide rail, on which a bottom face or a side face of the waste liquid box **60** may abut, and the guide rail may restrict the waste liquid box **60** from moving further inward.
(25) The waste liquid box exchangeable unit **15** may not necessarily have the cover **151** being openable/closable. For example, a waste liquid box unit, which consists of the waste liquid box **60** and a box retainer having the cover **151** to retain the waste liquid box **60**, may be mounted in and demounted from the main housing **11**. In this arrangement, when the waste liquid box unit is demounted from the main housing **11**, an opening having a form that corresponds to a form of the waste liquid box unit may be exposed; and when the waste liquid box unit is mounted in the main housing **11**, the cover **151** attached to the waste liquid box unit may close the opening.
(26) The power unit **16** includes a connector **161** on the main housing **11** and a power cord **162**. The power cord **162** is connected to the connector **161** at one end and, when connected to the connector **161** at the one end, extends outward from the main housing **11**. The power cord **162** has a plug, which may be plugged into an external power outlet, on an end opposite to the one end to be connected to the connector **161**. The power unit **16** may be supplied with power to the inkjet printer **1** from the external outlet. The connector **161** may be located at a rearward position on the sideward face of the main housing **11**. Optionally, the connector **161** may be located at a corner of the sideward face and a rearward face of the main housing **11** or may be located on the rearward face of the main housing **11**.
(27) Optionally, moreover, the power cord **162** may be detachably connected to the connector **161** or may be connected fixedly to the connector **161**. In the arrangement where the power cord **162** is detachable from the connector **161**, the connector **161** may serve as an inlet, and the end of the power cord **162** opposite to the outlet plug may have an inlet plug.
(28) The gripper **171** may consist of a recess formed on the sideward face of the main housing **11**. The gripper **171** allows a user to grab there-onto in order to lift or move the inkjet printer **1**. Optionally, the gripper **17** may be formed to protrude from the sideward face of the main housing **11**. The gripper **17** may be located at a position to coincide with a gravity center of the inkjet

printer **1** in the front-rear direction and in the vertical direction. In other words, the gripper **17** may be located at a substantially central position in the front-rear direction and the vertical direction on the sideward surface of the main housing **11**.
(29) The opening of the mountable compartment **152**, which may be exposed when the cover **151** located on the sideward face of the main housing **11** on the widthwise end is open, is located frontward with respect to the connector **161**, through which the power cord **162** is connected to the main housing **11**. In the present embodiment, a frontward side is a side, on which the opening **20** is formed, and a side, toward which the sheet P is ejected. The sheet P may be, as will be described further below, conveyed from the feeder tray **21** to the ejection tray **22** and ejected outward from the ejection tray **22**. The opening of the mountable compartment **152** is located frontward with respect to the gripper **17**.
(30) The opening of the mountable compartment **152** is located between a first position X**1**, which is apart by a predetermined distance from an edge where the rearward face of the main housing **11** and the sideward face of the main housing **11** meet, and a second position X**2**, which is apart by a predetermined distance from an edge where the frontward face of the main housing **11** and the sideward face of the main housing **11** meet. Therefore, the opening of the mountable compartment **152** is located apart from corners of the main housing **11** by predetermined distances. In other words, the opening of the mountable compartment **152** is located at a position excluding the corners. A distance between the first position X**1** and the second position X**2**, which is a dimension of the opening of the mountable compartment **152** in the front-rear direction, is greater than a dimension of the waste liquid box **60** in the front-rear direction.
(31) <Internal Configuration>
(32) Next, an internal configuration of the inkjet printer **1** will be described with reference to FIG. **2**.
(33) As shown in FIG. **2**, the inkjet printer **1** includes a feeder **3**, a recorder **4**, a conveyer and a controller **100**.
(34) The feeder **3** includes a shaft **30**, a feeder arm **31**, and a feeder roller **32**. The feeder **3** may feed sheets P contained in the feeder tray **21** to a conveyer path R by forwarding rotation of the feeder roller **32**. The feeder roller **32** is located at a tip end of the feeder arm **31** and is supported rotatably by the feeder arm **31**. The feeder arm **31** is pivotably supported by the shaft which is supported by a frame of the inkjet printer **1**. The feeder arm **31** is urged toward the feeder tray **21** by weight thereof or by an urging force from, for example, a spring.
(35) The conveyer path R extends upward from a rear end of the feeder tray **21**, curving frontward in an area delimited by a guide member **33**, to the ejection tray **22**.
(36) The feeder roller **32** may, when a feeder motor (not shown) is activated by the controller **100**, pick up the sheets P from the feeder tray **21** one by one. The sheets P picked up from the feeder tray **21** may be conveyed along the conveyer path R and fed to the recorder **4**.
(37) The recorder **4** is located above the feeder **3**. The recorder **4** includes the carriage **40**, a recording head **41** for discharging liquid, a plurality of nozzles **42**, and a platen **43**. The carriage **40** is supported by guide rail **9**A and a guide rail **9**B, which extend in the widthwise direction (see FIG. **3**). The carriage **40** may, when a driving force from a carriage motor (not shown) is transmitted thereto, move back and forth in the scanning direction which is the widthwise direction, i.e., a direction of width of the sheet P being conveyed. For recording an image on the sheet P, the controller **100** of the inkjet printer **1** may repeat a recording process, in which the controller **100** operates the carriage **40** to move in the widthwise direction and the recording head **41** to discharge ink through the nozzles **42** while the sheet P stays still, and a conveying process, in which the controller **100** drives the conveyer roller **50** and the ejection roller **52** to convey the sheet P by a predetermined linefeed amount, alternately.
(38) On the carriage **40**, the recording head **41** is mounted. The plurality of nozzles **42** are formed on a lower face of the recording head **41**. The plurality of nozzles **42** are arrayed in lines along the

front-rear direction to form nozzle lines, and four (4) nozzle lines are formed on a nozzle surface **421** to align in the widthwise direction. The nozzles **42** forming a first one of the nozzle lines, a second one of the nozzle lines, a third one of the nozzle lines, and a fourth one of the nozzle lines from right to left, may discharge inks in colors of black, yellow, cyan, and magenta, respectively. However, the aligning order of the nozzle lines may not necessarily be limited but may be changed optionally, for example, on a product model basis.
(39) The recording head **41** may discharge liquid, i.e., the inks, through the nozzles **42** by causing vibrating elements such as piezo elements to vibrate.
(40) The platen **43** is located at a position below the recording head **41** and extends throughout or over the entire length of the sheet P in the widthwise direction. The platen **43** may support the sheet P from below during the recording process. While the carriage **40** moves in the widthwise direction over the sheet P supported by the platen **43**, the recording head **41** may discharge the ink droplets selectively from the nozzles **42** to record the image on the sheet P.
(41) The conveyer **5** includes the conveyer roller **50** and the ejection roller **52**, which are located on one side and the other side, respectively, of the carriage **40** and the platen **43** in the front-rear direction. At a position below the conveyer roller **50**, a pinch roller **51** is arranged to face the conveyer roller **50**. The conveyer roller **50** may be driven by a conveyer motor (not shown) to rotate. The pinch roller **51** may rotate along with the rotation of the conveyer roller **50**. With the rotation of the conveyer roller **50** and the pinch roller **51**, the sheet P nipped between the conveyer roller **50** and the pinch roller **51** may be conveyed along the conveyer path R to the recorder **4**.
(42) The ejection roller **52** is located on a downstream side of the conveyer roller **50** across the carriage **40** and the platen **43** in the conveying direction. At a position above the ejection roller **52**, a spur roller **53** is arranged to face the ejection roller **52**. The ejection roller **52** may be driven by the conveyer motor (not shown) to rotate. The spur roller **53** may rotate along with the rotation of the ejection roller **52**. With the rotation of the ejection roller **52** and the spur roller **53**, the sheet P nipped between the ejection roller **52** and the spur roller **53** may be ejected from the conveyer path R to rest at the ejection tray **22**.
(43) The controller **100** includes a Central Processing Unit (CPU), a Read-Only Memory (ROM), a Random Access Memory (RAM), and Application Specific Integrated Circuit (ASIC) including a variety of controlling circuits. The controller **100** is connected with devices that compose the inkjet printer **1**, including the recording head **41** and the conveyer motor of the conveyer **5**. The controller **100** is, moreover, connected with the operation panel **13** and external devices such as a PC (not shown).
(44) The controller **100** may run programs stored in the ROM to cause the CPU and the ASIC to execute processes to control acts of the devices, including the recording head **41** and a maintenance unit **81**, such as a process called flushing to discharge ink droplets from the nozzles **42**. The controller **100** may control the recording head **41** and the conveyer motor according to a printing command transmitted from the external device such as the PC and execute a printing process to print the image on the sheet P. The maintenance unit **81** and the flushing process will be described further below. It may be noted that the controller **100** may not necessarily consist of the CPU, the ROM, the RAM, and the ASIC alone but may consist of any hardware devices.
(45) The inkjet printer **1** in the configuration as described above may operate the conveyer **5** to convey the sheet P in the conveying direction, move the carriage **40** along with the recording head **41** in the scanning direction, and operate the recording head **41** to discharge the ink to print an image on the sheet P.
(46) <Cartridge Case **140**>
(47) Next, the cartridge case **140** will be described with reference to FIG. **3**. As shown in FIG. **3**, the cartridge case **140** may contain four (4) ink cartridges **141** aligning in the widthwise direction. The ink cartridges **141** are removable from the cartridge case **140** and may contain the colored inks to be supplied to the recording head **41**. For example, the ink cartridges **141** may contain the inks in

the colors of black, yellow, cyan, and magenta. The inks contained in the ink cartridges **141** may be supplied to the recording head **41** through tubes **142**.
(48) <Maintenance Unit **81**>
(49) Next, the maintenance unit **81** will be described with reference to FIG. **3**. As shown in FIG. **3**, the inkjet printer **1** has the maintenance unit **81**. The maintenance unit **81** includes a cap **811**, a pump **812**, and a tube **813**. The maintenance unit **81** is located at a position lower than a scanning path of the carriage **40** and on an outer side from the platen **43** in the scanning direction. In the present embodiment, the maintenance unit **81** is located rightward with respect to the platen **43**. However, the maintenance unit **81** may optionally be located leftward with respect to the platen **43**.
(50) The cap **811** is formed of rubber. The cap **811** is located at a position lower than the recording head **41** and on the outer side from the platen **43** in the scanning direction. When the carriage **40** is located at a maintenance position, which is on the outer side from the platen **43** in the scanning direction, the cap **811** faces the nozzle surface **421** of the recording head **41**.
(51) The cap **811** is movable in the vertical direction by, for example, a lifting device (not shown). The cap **811** located at the maintenance position may move upward to fit with the nozzle surface **421**. The cap **811** may collect the ink discharged from the recording head **41**. The cap **811** fitting with the nozzle surface **421** may cover the recording head **41** to receive the inks discharged from the recording head **41** and prevent the nozzle surface **421** from drying.
(52) The pump **812** may be driven by a motor (not shown) to suction the inks in the nozzles **42** through the cap **811** and the tube **813** and eject the suctioned inks through the tube **813** at the waste liquid box **60**.
(53) The tube **813** forms a flow path, in which the inks ejected from the maintenance unit **81** may be transported to the waste liquid box **60**. The tube **813** is made of a flexible material. The inks may be transported from the cap **811** through the pump **812** to the waste liquid box **60**.
(54) The maintenance unit **81** in the configuration as described above may perform maintenance works on the recording head **41**. In particular, the maintenance unit **81** may perform a purging action to suction the inks and the air in the nozzles **42** and dust adhered to the nozzles **42** as a maintenance work. In the following paragraphs, the term “inks” may include the inks and the air in the nozzles **42** and the dust adhered to the nozzles **42**. The inks removed from the recording head **41** by the maintenance unit **81** may be stored in the waste liquid box **60**.
(55) <Flushing Unit **70**>
(56) The flushing unit **70** may prevent the inks that are discharged from the nozzles **42** from scattering in mist and guide the inks to the waste liquid box **60** during a flushing process. At a position below the flushing unit **70**, optionally, an intermediate box (not shown) may be provided. The intermediate box may have an opening formed on an upper side at a position to coincide with the nozzles **42**, and the inks discharged in the flushing process may be guided through the intermediate box to the waste liquid box **60**. Optionally, the flushing unit **70** may be configured to guide the inks discharged in the flushing process directly to the waste liquid box **60**.
(57) The flushing process is an action to operate the recording head **41** while the carriage **40** is located at a flushing position, where the carriage **40** faces the flushing unit **70**, to discharge the inks through the nozzles **42** at the flushing unit **70**, rather than at the sheet P.
(58) In the present embodiment, the maintenance unit **81** and the flushing unit **70** are located on outer sides of the platen **43** in the scanning direction. In particular, the flushing unit **70** is located on the opposite side to the maintenance unit **81** across the conveyer path R in the widthwise direction. For example, the maintenance unit **81** may be located on the rightward side of the conveyer path R, and the flushing unit **70** may be located on the leftward side of the conveyer path R. However, the arrangement of the maintenance unit **81** and the flushing unit **70** may not necessarily be limited to the arrangement as described above or shown in FIG. **3**.
(59) <Waste Liquid Box **60**>
(60) The waste liquid box **60** is a storage for storing the inks discharged from the recording head

41. In particular, the waste liquid box **60** may store the inks discharged from the recording head **41** during the maintenance work and the flushing process. The waste liquid box is demountably mounted in the main housing **11**.
(61) The waste liquid box **60** is mountable in and demountable from the main housing **11** through the opening of the mountable compartment **152**, which may be exposed when the cover **151** is open. According to the present embodiment, the mountable/demountable direction of the waste liquid box **60** is the widthwise direction and is different from the ejecting direction, i.e., the front-rear direction, to eject the sheet P outward through the ejection tray **22**. The waste liquid box **60** is mountable in and demountable from the main housing **11** through the opening of the mountable compartment **152** located frontward with respect to the connector **161**, through which the power cord **162** is connected to the main housing **11**, in the front-rear direction.
(62) Arrangement of the waste liquid box **60** will be described with reference to FIGS. **3** and **4**. FIG. **4** illustrates the arrangement of the waste liquid box **60** viewed from the upper side of the inkjet printer **1**.
(63) As shown in FIGS. **3** and **4**, the waste liquid box **60** is located at a position, at which the waste liquid box **60** when mounted in the main housing **11** is at least partly lower than the feeder tray **21** and the ejection tray **22** and to at least partly overlap the feeder tray **21** and the ejection tray **22** in the vertical direction. In other words, as shown in FIG. **4**, at least a part of the waste liquid box **60** is located to coincide with the feeder tray **21** and the ejection tray **22** in the vertical direction.
(64) The ink cartridges **141** are arranged alongside the waste liquid box **60**, the feeder tray **21**, and the ejection tray **22** in the widthwise direction. For example, as shown in FIG. **3**, the ink cartridges **141** may be located rightward, and the waste liquid box **60**, the feeder tray **21**, and the ejection tray **22** may be located leftward. Optionally, for another example, the ink cartridges **141** may be located leftward, and the waste liquid box **60**, the feeder tray **21**, and the ejection tray **22** may be located rightward. Meanwhile, the opening of the mountable compartment **152**, in which the waste liquid box **60** is mountable, is located on the sideward face of the main housing **11** on an opposite side to the ink cartridges **141** across the conveyer path R for the sheet P. In other words, the waste liquid box **60** may be mountable in and demountable from the main housing **11** through the sideward face, which is located on the opposite side to the ink cartridges **141** across the conveyer path R for the sheet P.
(65) In this arrangement, the waste liquid box **60** and the ink cartridges **141** may not necessarily align at a same vertical position. In other words, for example, the ink cartridges **141** may be located to be higher than the feeder tray **21** and the ejection tray **22** while the waste liquid box **60** may be located to be lower than the feeder tray **21** and the ejection tray **22**. A dimension of the waste liquid box **60** in the widthwise direction is greater than dimensions of the feeder tray **21** and the ejection tray **22** in the widthwise direction. In the arrangement where the waste liquid box **60** and the ink cartridges **141** align side by side in the widthwise direction, a rightward end of the waste liquid box **60** may be located at a position to reach a leftward end of the cartridge case **140**, in which the ink cartridges **141** are accommodated. In other words, the dimension of the waste liquid box **60** in the widthwise direction, i.e., a dimension of the waste liquid box **60** in a lengthwise direction, may be at most equal to a distance between the leftward end of the cartridge case **140** and the opening of the mountable compartment **152**.
Benefits
(66) According to the configuration described above, while the direction to eject the sheet P from the main housing **11** coincides with the front-rear direction, the waste liquid box **60**, which is mountable through the opening of the mountable compartment **152** located at the frontward position on the sideward face of the main housing **11** on the widthwise end, may be mountable in and demountable from the main housing **11** in the direction different from the ejecting direction. Therefore, when, for example, the sheet P jams the ejection tray **22** while the inkjet printer **1** prints images on a large amount of sheets P or on a longer sheet P, the ejection tray **22** or the jammed

sheet P may not interfere with the waste liquid box **60** to be demounted from or mounted in the main housing **11**, and the waste liquid boxes **60** may be easily exchanged. For another example, when a longer-sized printing sheet is set to droop down on the front face of the main housing **11**, the waste liquid box **60** may be demounted from or mounted in the main housing **11** without interfering with the printing sheet drooping down on the front face of the main housing **11**. Therefore, a user may not be urged to remove the longer-sized printing sheet for exchanging the waste liquid boxes **60**. For another example, according to the configuration described above, the waste liquid box **60** is mountable in or demountable from the main housing **11** through the frontward area on the sideward face of the main housing **11**; therefore, while, for example, the inkjet printer **1** is located in an arrangement such that a rearward face of the main housing **11** faces toward a wall, the user may access the waste liquid box **60** easily for demounting and mounting. Moreover, the waste liquid box **60** is mountable in and demountable from the main housing **11** through the frontward area on the sideward face of the main housing **11** with respect to the connector **161**, through which the power cord **162** is connected to the main housing **11**. Therefore, the waste liquid boxes **60** may be exchanged smoothly without interfering with the power cord **163**.

(67) According to the configuration described above, moreover, the waste liquid box **60** is mountable in and demountable from the main housing **11** through the opening of the mountable compartment **152** located frontward with respect to the gripper **17**. Therefore, the waste liquid boxes **60** may be exchanged easily.

(68) Further, according to the arrangement of the waste liquid box **60** described above, the mountable compartment **152** is open at the position, which is displaced from the corners of the main housing **11**. When an opening is formed in the vicinity of a corner of the main housing **11**, strength of the main housing **11** tends to be lowered. In this regard, according to the present embodiment, the opening of the mountable compartment **152** is located at the position apart from the corners of the main housing **11** by predetermined distances. Therefore, the strength of the main housing **11** may be restrained from lowering.

(69) According to the configuration described above, moreover, the liquid, which was discharged from the recording head **41** but was not used in recording the image on the sheet P, may be collected by the waste liquid collecting devices and stored in the waste liquid box **60**. The waste liquid collecting devices include, for example, the cap **811** used in the maintenance work with the recording head **41**. Therefore, the inks suctioned from the nozzles **42** of the recording head **41** in the maintenance work may be collected and stored in the waste liquid box **60**. Moreover, the inkjet printer **1** may collect the inks discharged from the nozzles **42** in the flushing process. Thus, the inks discharged from the recording head **41** in the flushing process may be collected and stored in the waste liquid box **60**.

(70) According to the configuration described above, moreover, the waste liquid box **60** is located at the position lower than the feeder tray **21** and the ejection tray **22** to overlap the feeder tray **21** and the ejection tray **22** in the vertical direction and to align side by side with the ink cartridges **141** in the widthwise direction. Therefore, the waste liquid box **60** may be expanded at the position lower than the feeder tray **21** and the ejection tray **22** to the vicinity of the ink cartridges **141**. In other words, without increasing the dimension of the main housing **11** in the widthwise direction, the waste liquid box **60** with larger capacity may be provided. With the waste liquid box **60** having the larger capacity, frequency to exchange the waste liquid boxes **60** may be lowered. Furthermore, a height of the ink cartridges **141** may be increased to a dimension of the waste liquid box **60** in the vertical direction. Therefore, the ink cartridges **141** with larger capacities may be provided.

(71) According to the configuration described above, moreover, the opening of the mountable compartment **152** is located on the sideward face of the main housing **11** on the side opposite to the ink cartridges **141** across the conveyer path R. Therefore, the waste liquid box **60** may be mounted in and demounted from the side opposite to the ink cartridges **141**. Accordingly, the waste liquid boxes **60** may be exchanged without interfering with the ink cartridges **141**.

(72) According to the configuration described above, moreover, the dimension of the waste liquid box **60** in the widthwise direction may be increased to be greater than the dimensions of the feeder tray **21** and the ejection tray **22**; therefore, the waste liquid box **60** with a larger capacity may be provided. Accordingly, frequency to exchange the waste liquid boxes may be lowered.

First Modified Example

(73) In the first embodiment described above, the waste liquid box **60** is located to align with the ink cartridges **141** in the widthwise direction. However, the waste liquid box **60** may not necessarily be located to align with the ink cartridges **141** in the widthwise direction. A first modified arrangement of the waste liquid box **60** will be described with reference to FIGS. **5** and **6**. FIG. **5** illustrates a modified arrangement of the waste liquid box **60** viewed from the upper side. FIG. **6** illustrates the modified arrangement of the waste liquid box **60** viewed from a front side.

(74) As shown in FIG. **5**, the waste liquid box **60** may be located rearward with respect to the ink cartridges **141**. In other words, the waste liquid box **60** located rearward with respect to the ink cartridges **141** may not overlap the ink cartridges **141** in the widthwise direction.

(75) According to the modified arrangement, the waste liquid box **60** being located rearward with respect to the ink cartridges **141** may be mounted in and demounted from the main housing **11** without interfering with the ink cartridges **141**. Therefore, the waste liquid boxes **60** may be exchanged easily.

(76) According to the modified arrangement, moreover, the dimension of the waste liquid box **60** in the widthwise direction may not be affected by the location of the ink cartridges **141**. Therefore, the waste liquid box **60** may be, as shown in FIGS. **5** and **6**, extended rightward to a position in the vicinity of the rightward end of the main housing **11**. In other words, the dimension of the waste liquid box **60** in the widthwise direction may be, at most, equal to a distance between an inner surface of the sideward wall of the main housing **11** on the right and the opening of the mountable compartment **152**. Therefore, without increasing the dimension of the main housing **11** in the widthwise direction, the waste liquid box **60** with a larger capacity may be provided.

Second Modified Example

(77) A second modified arrangement of the waste liquid box **60** will be described with reference to FIG. **7**. FIG. **7** illustrates the second modified arrangement of the waste liquid box viewed from the upper side.

(78) As shown in FIG. **7**, the waste liquid box **60** may be mounted in the mountable compartment **152** in a posture such that an edge of the frontward face and toward the opening of the mountable compartment **152**, e.g., the leftward-front edge, is located frontward with respect to an edge of the frontward face and opposite to the opening of the mountable compartment **152**, e.g., the rightward-front edge, in the front-rear direction. In other words, the waste liquid box **60** may be mounted in the mountable compartment **152** in a skewed posture, in which the edge of the frontward face and toward the opening of the mountable compartment **152** is located frontward, and the edge of the frontward face and opposite to the opening of the mountable compartment **152** is located rearward. In this arrangement, the mountable compartment **152** may have walls that are skewed with respect to the front face of the main housing **11** in the top plan view inside the main housing **11**. Therefore, a user may insert the waste liquid box **60** in the mountable compartment **152** to abut on one of the walls and slide along the wall inward to mount the waste liquid box **60** in the skewed posture inside the main housing **11**. Moreover, optionally, the mountable compartment **152** may have a guide rail in a skewed posture with respect to the front face of the main housing **11** in the top plan view to guide the waste liquid box **60**.

(79) According to the second modified arrangement, the waste liquid box **60** may be mounted in or demounted from the main housing **11** from a frontward oblique position. Therefore, for example, when an obstacle object is located on the widthwise side of the main housing **11**, while it may be difficult to mount or demount the waste liquid box **60** in or from the main housing **11** in the widthwise direction, the waste liquid box **60** may be mounted in or demounted from the main

housing **11** in the skewed direction easily. Therefore, the waste liquid boxes **60** may be exchanged easily.
Third Modified Example
(80) A third modified arrangement of the waste liquid box **60** will be described with reference to FIG. **8**. FIG. **8** illustrates the third modified arrangement of the waste liquid box **60** viewed from the upper side.
(81) As shown in FIG. **8**, the waste liquid box **60** may be at least partly located rearward with respect to the position to locate the ink cartridges **141** in the main housing **11**. For example, as shown in FIG. **8**, the waste liquid box **60** may have a form of an L in the upper plan view, and a part of the L form may be located rearward with respect to the ink cartridges **141**, and the other part of the L form may be located sideward in the widthwise direction with respect to the ink cartridges **141**.
(82) Thus, a dimension of a part of the waste liquid box **60** that overlaps the ink cartridges **141** in the widthwise direction may be reduced by a dimension of the ink cartridges **141** in the widthwise direction, while a dimension of a remainder of the waste liquid box **60** may be greater. According to the third modified arrangement of the waste liquid box **60**, the ink cartridges **141** may be prevented from interfering with the waste liquid box **60** being mounted in or demounted from the main housing **11**. Therefore, the waste liquid boxes **60** may be exchanged easily, and the waste liquid box **60** with a larger capacity may be provided.
Second Embodiment
(83) A second embodiment of the present disclosure will be described below. In the following paragraphs, items that are substantially identical to those described above will be referred to by the same reference signs, and description of those items is herein omitted.
(84) An inkjet printer **200** according to the second embodiment will be described with reference to FIGS. **9**, **10**, and **11**A-**11**B. FIG. **9** is an illustrative view of the inkjet printer **200** viewed from a front side. FIG. **10** is an illustrative view of the inkjet printer **200** viewed from an upper side. FIGS. **11**A-**11**B are illustrative views of an intermediate box **700**, the waste liquid box **60**, and a waste liquid guide **90** in the inkjet printer **200**.
(85) The inkjet printer **200** has the maintenance unit **81**, a wiper **82**, the platen **43**, a duct **86**, and the waste liquid guide **90** for collecting the waste liquid. The inkjet printer **200** has an intermediate box **700** with an outlet **75**, through which the inks collected by the maintenance unit **81**, the wiper **82**, the platen **43**, the duct **86**, and the waste liquid guide **90** may be ejected into the waste liquid box **60**.
(86) <Configurations of Waste Liquid Collecting Devices>
(87) Next, with reference to FIGS. **9-10**, waste liquid collecting devices will be described. The arrangement of the items in the inkjet printer **200** shown in FIGS. **9-10** is merely an illustrative example and, unless otherwise noted, may not necessarily limit arrangement of the other items in the inkjet printer **200**.
(88) As shown in FIGS. **9-10**, the inkjet printer **200** has the recording head **41**, the platen **43**, the maintenance unit **81**, the wiper **82**, a liquid reservoir **83**, a cleaning liquid injector unit **84**, the duct **86**, the intermediate box **700**, the waste liquid box **60**, and the waste liquid guide **90**.
(89) The platen **43** may collect the inks discharged on the platen **43**. The platen **43** has a slant surface, which slants in the front-rear direction to be lower toward the rear side. As shown in FIGS. **9-10**, on a surface of the platen **43** toward the nozzles **42**, a plurality of ribs **46** elongated in the front-rear direction are arranged. The platen **43** may support the sheet P on the plurality of ribs **46**. As shown in FIG. **10**, the platen **43** has a waste liquid chute **45** to collect the inks, i.e., waste liquid, flowing down along the slant surface of the platen **43**. The waste liquid chute **45** may consist of, for example, gutters or grooves.
(90) The waste liquid chute **45** is arranged in a rearend area and a leftward area in the platen **43**, approximately in a form of L. In the following description, the rearward part of the waste liquid

chute **45** will be called a rearward waste liquid chute **451**, and the leftward part of the waste liquid chute **45** will be called a sideward waste liquid chute **452**. In other words, the rearward waste liquid chute **451** and the sideward waste liquid chute **452** will be collectively called “waste liquid chute **45**.” The rearward waste liquid chute **451** slants to be lower toward the left. In the sideward waste liquid chute **452**, a liquid outlet **44** is formed. The sideward waste liquid chute **452** slants to be lower toward the liquid outlet **44**. Thus, the waste liquid chute **45** has a form, in which the collected inks may flow down toward the liquid outlet **44**.
(91) The inks discharged on the platen **43** may flow rearward on the platen **43** along the slant surface in a direction indicated by an arrow Y**1**. The rearward waste liquid chute **451** may receive the inks flown in the direction of the arrow Y**1**. The inks flowing in the rearward waste liquid chute **451** may flow in a direction indicated by an arrow Y**2** to the sideward waste liquid chute **452**. The inks reaching the sideward waste liquid chute **452** may flow in a direction indicated by an arrow Y**3** to the liquid outlet **44**.
(92) As shown in FIG. **9**, the intermediate box **700** is located at a position leftward with respect to the platen **43** and lower than the platen **43**. The intermediate box **700** is located below the liquid outlet **44**. As indicated by an arrow Y**9** in FIG. **9**, the inks flowing in the sideward waste liquid chute **452** may be drained from the platen **43** through the liquid outlet **44** and flow into the intermediate box **700**. As shown in FIG. **10**, the liquid outlet **44** is located at a position to overlap an opening **72** of the intermediate box **700**, which will be described further below, in a view along the vertical direction. Therefore, the inks dripping down from the liquid outlet **44** may fall through the opening **72** of the intermediate box **700** and may be stored in the intermediate box **700**.
(93) The maintenance unit **81** may restore a discharging condition of the nozzles **42** in the recording head **41** to an earlier condition. As shown in FIG. **10**, the maintenance unit **81** is located on an outer side of the platen **43** in the widthwise direction. The maintenance unit **81** is located on a same side of the platen **43** as a standby position of the recording head **41**. The standby position of the recording head **41** is a position, at which the carriage **40** stands by when the inkjet printer **200** does not print an image on the sheet P. For example, the maintenance unit **81** may be located rightward with respect to the platen **43**. The maintenance unit **81** is located at a position lower than the recording head **41** within a scanning range of the recording head **41**. The maintenance unit **81** has a cap (not shown) and a suction pump (not shown).
(94) The cap is located at a position to face the recording head **41** located at the standby position. The cap may collect the inks being waste liquid from the nozzles **42**. The cap may be driven to move up and down by power of a driving motor (not shown). The cap may move upward to a position where the cap covers the nozzle surface **421** of the recording head **41** located at the standby position. The cap covering the nozzle surface **421** may seal the nozzle surface **421**.
(95) The suction pump is connected to the cap. The suction pump may be driven by the driving motor to suction the inks being the waste liquid from the nozzles **42** sealed by the cap. In other words, the suction pump may collect the inks from the nozzles **42**. The driving motor may be controlled by the controller **100**. The suction pump may be, for example, a tube pump. The suction pump is connected with a waste liquid tube **55**. The inks suctioned from the nozzles **42** by the suction pump may flow in a direction of an arrow Y**4** through the waste liquid tube **55** and may be ejected through a port **711** into the intermediate box **700**.
(96) The wiper **82** may wipe the nozzle surface **421** of the recording head **41** to remove the inks from the nozzles **42**. In other words, the wiper **82** may collect the inks adhered to the nozzles **42** by wiping the nozzle surface **421**. The wiper **82** may be made of rubber. The wiper **82** is located at a position in the scanning range of the recording head **41**. The wiper **82** is located to be lower than the recording head **41**. The wiper **82** is located between the platen **43** and the maintenance unit **81** in the widthwise direction.
(97) An upper end of the wiper **82** may contact the nozzle surface **421** of the recording head **41**. The wiper **82** is supported by a holder (not shown) and may move in the vertical direction. When

the inkjet printer **200** records an image, the wiper **82** moves downward to a position where the wiper **82** may not contact the nozzle surface **421**. After the maintenance unit **81** suctions the ink from the nozzles **42**, the wiper **82** may move upward to a position where the wiper **82** contacts the nozzle surface **421**. The wiper **82** moved to the upper position may contact the nozzle surface **421** of the recording head **41** that runs on the wiper **82**. Thereby, the wiper **82** may wipe the nozzle surface **421** and remove the inks being the waste liquid adhered to the nozzles **42**. The inks removed from the nozzles **42** by the wiper **82** may flow on the wiper **82** into the liquid reservoir **83**.
(98) The cleaning liquid injector unit **84** is located in the vicinity of the wiper **82**. The cleaning liquid injector unit **84** is located on the same side of the platen **43** as the wiper **82** in the scanning direction of the recording head **41**. The cleaning liquid injector unit **84** includes a cleaning liquid injector nozzle **841**, a cleaning liquid supplying tube **842**, and a cleaning liquid reservoir **843**.
(99) The cleaning liquid injector nozzle **841** may inject cleaning liquid for cleaning at the wiper **82**. The cleaning liquid injector nozzle **841** is located at a position where the cleaning liquid from the cleaning liquid injector nozzle **841** may reach the wiper **82**. The cleaning liquid is stored in the cleaning liquid reservoir **843**. When the controller **100** operates a cleaning liquid supplying unit (not shown), which includes a pump, the cleaning liquid supplying unit may supply the cleaning liquid stored in the cleaning liquid reservoir **843** to the cleaning liquid injector nozzle **841** through the cleaning liquid supplying tube **842**. The cleaning liquid injector nozzle **841** may inject the cleaning liquid supplied from the cleaning liquid reservoir **843** at the wiper **82**. The cleaning liquid injected from the cleaning liquid injector nozzle **841** may be stored in the liquid reservoir **83** as waste liquid.
(100) As shown in FIG. **9**, the liquid reservoir **83** is located below the wiper **82**. The liquid reservoir **83** may store the ink removed by the wiper **82** from the nozzle surface **421** and the cleaning liquid injected from the cleaning liquid injector nozzle **841**. In other words, the liquid reservoir **83** may collect the cleaning liquid injected from the cleaning liquid injector nozzle **841**.
(101) The liquid reservoir **83** is a casing having an opening **831** on an upper side thereof. The ink removed by the wiper **82** and the cleaning liquid injected from the cleaning liquid injector nozzle **841** may fall through the opening **831** to be stored in the liquid reservoir **83**. The liquid reservoir **83** is connected with the intermediate box **700** through a waste liquid tube **55**. The inks and the cleaning liquid collected in the liquid reservoir **83** may flow into the waste liquid tube **55**, as indicated by an arrow Y**5** shown in FIGS. **9** and **10**. Further, the inks and the cleaning liquid may be ejected through the waste liquid tube **55** and a port **712** into the intermediate box **700**.
(102) The duct **86** may suction mist of liquid produced in the main housing **11**. In particular, the duct **86** may suction mist of the inks discharged from the nozzles **42** of the recording head **41** and/or mist of the cleaning liquid injected from the cleaning liquid injector nozzle **841**. The duct **86** is located on a leftward side in the main housing **11**. The duct **86** includes a fan **89** and a conduit **87**.
(103) The fan **89** may expel the air inside the main housing **11** to the outside of the main housing **11** through an air outlet **891**. The fan **89** may be driven by the controller **100**. The fan **89** is connected with the conduit **87** being a passage for the air. The conduit **87** is a passage, through which the air suctioned from the inside of the main housing **11** may be transported to the air outlet **891**. One end of the conduit **87** is connected to the fan **89** and to the air outlet **891**, and the other end of the conduit **87** is connected to an air inlet **88**, through which the air inside the main housing **11** may be drawn into the conduit **87**.
(104) When the fan **89** is driven, the air pressure inside the main housing **11** may be lowered, and the air in the main housing **11** may be suctioned through the air inlet **88**. The liquid in the form of mist discharged from the nozzles **42** of the recording head **41** may be suctioned together with the air inside the main housing **11** through the air inlet **88**, as indicated by an arrow Y**7** shown in FIG. **9**. The liquid in the form of mist may be condensed in the conduit **87** into liquid and accumulate as waste liquid. To the conduit **87**, a waste liquid tube **55** is connected. The waste liquid accumulated

in the conduit **87** may be ejected through the waste liquid tube **55** and a port **713** into the intermediate box **700**, as indicated by an arrow Y**8** in FIG. **10**.
(105) The controller **100** may execute the flushing process, in which the controller **100** operates the recording head **41** to discharge the inks from the nozzles **42** at the opening **72** of the intermediate box **700**. For executing the flushing process, the controller **100** may move the recording head **41** to a flushing area, where the flushing process is performed.
(106) In FIGS. **9** and **10**, the recording head **41** located in the flushing area is illustrated. The flushing process is performed in the flushing area, which is on an opposite side to the standby position of the recording head **41** across the platen **43**. In the present embodiment, the flushing process is executed on a leftward side to the platen **43** in the widthwise direction. The flushing area is an area above the opening **72** of the intermediate box **700**. As indicated by an arrow Y**10** shown in FIG. **9**, the inks discharged from the nozzles **42** during the flushing process, i.e., waste liquid, may fall through the opening **72** to be stored in the intermediate box **700**.
(107) As shown in FIG. **9**, in a direction of height of the inkjet printer **200**, i.e., in the vertical direction, at a position above the intermediate box **700**, the waste liquid guide **90** is arranged. The waste liquid guide **90** may receive the inks discharged from the recording head **41** in the flushing process. The waste liquid guide **90** is a piece that may guide the inks discharged from the nozzles **42** in the flushing process to the opening **72** of the intermediate box **700**. The waste liquid guide **90** is, in the direction of height of the inkjet printer **200**, located to be lower than the recording head **41**. The waste liquid guide **90** is located between the nozzles **42** of the recording head **41** located in the flushing area and the opening **72** of the intermediate box **700**. The waste liquid guide **90** is retained by a retainer, which is not shown. A number of piece(s) of the waste liquid guide **90** may not necessarily be limited as long as at least one piece of waste liquid guide **90** is provided.
(108) The waste liquid guide **90** and the intermediate box **700** will be described below with reference to FIGS. **11**A-**11**B. FIG. **11**A illustrates a cross-sectional view of the intermediate box **700**, the waste liquid box **60**, and the waste liquid guide **90**. FIG. **11**B illustrates a top plan view of the intermediate box **700** and the waste liquid guide **90**. In FIG. **11**A, illustration of the ports **711**, **712**, **713** in the intermediate box **700** is omitted for a viewer's easier understanding.
(109) As shown in FIG. **11**A, an end **92** of the waste liquid guide **90** toward the recording head **41** and an end **91** of the waste liquid guide **90** toward the intermediate box **700** are located to be higher than the opening **72**. Optionally, however, the waste liquid guide **90** may be arranged to partly overlap the opening **72** in the vertical direction. A surface of the waste liquid guide **90** on a side toward the nozzles **42** is a slant surface **93**, which slants with respect to the opening **72**.
(110) FIG. **11**B illustrates a positional relation between the waste liquid guide **90** and the opening **72** of the intermediate box **700**. In the top plan view in FIG. **11**B, for a viewer's easier understanding, a single piece of waste liquid guide **90** alone is illustrated while the other pieces of waste liquid guide **90** are omitted. In the top plan view to look down the main housing **11** from an upper side, the end **91** of the waste liquid guide **90** toward the intermediate box **700** is located to overlap the opening **72**.
(111) The inks discharged in the flushing process may land on the slant surface **93** of the waste liquid guide **90**. As indicted by an arrow X in FIG. **11**B, the inks landing on the slant surface **93** may flow down on the slant surface **93** and drip down from the end **91** of the waste liquid guide **90** to the intermediate box **700** through the opening **72**.
(112) <Configuration of Intermediate Box **700**>
(113) The intermediate box **700** is a tank to store the inks drained from a plurality of waste liquid collecting devices. The intermediate box **700** intervenes in flows of the inks between the waste liquid collecting devices and the waste liquid box **60**. In other words, the intermediate box **700** relays the inks to flow from the waste liquid collecting devices to the waste liquid box **60**. The intermediate box **700** has the ports **711**, **712**, **713**, the opening **72**, a first waste liquid absorber **73**, the outlet **75**, and a first waste liquid reservoir **78**.

(114) The ports **711**, **712**, **713** are parts of the intermediate box **700**, to which the waste liquid tubes **55** are connected. The opening **72** is formed on the upper face of the intermediate box **700**. The opening **72** of the intermediate box **700** is formed at a position coincident with the liquid outlet **44** in the platen **43** in the vertical direction. The opening **72** of the intermediate box **700** may accept the inks ejected from the liquid outlet **44** in the platen **43**. A size of the opening **72** of the intermediate box **700** is larger than a size of the liquid outlet **44** in the platen **43**. The opening **72** of the intermediate box **700** is formed at a position to coincide with the nozzles **42** of the recording head **41** in the vertical direction, and the opening **72** may accept the inks discharged from the nozzles **42**. The size of the opening **72** is larger than a size of the nozzle surface **421**.
(115) The liquid received through the ports **711**, **712**, **713** and the opening **72** may be stored in the first waste liquid reservoir **78**. In other words, the first waste liquid reservoir **78** may store the ink and the cleaning liquid drained from the platen **43**, the maintenance unit **81**, the liquid reservoir **83**, and the duct **86**. Moreover, the first waste liquid reservoir **78** may store the ink discharged in the flushing process.
(116) As shown in FIGS. **11**A-**11**B, the intermediate box **700** has the first waste liquid absorber **73** inside the first waste liquid reservoir **78**. The first waste liquid absorber **73** may absorb the liquid received in the first waste liquid reservoir **78**. The first waste liquid absorber **73** may be made of, for example, unwoven fabric, sponge, or cotton that may absorb the liquid. The first waste liquid absorber **73** has a protrusion **74** protruding toward the waste liquid box **60**. In particular, the protrusion **74** protrudes outward from the intermediate box **700** through the outlet **75** formed in the intermediate box **700**. In other words, the protrusion **74** is exposed outside the intermediate box **700** through the outlet **75**. Through the protrusion **74**, the liquid collected in the first waste liquid reservoir **78** may be drained outside the intermediate box **700**.
(117) <Configuration of Waste Liquid Box **60**>
(118) The waste liquid box **60** is mountable in and demountable from the main housing **11** through the opening of the mountable compartment **152**, which may be exposed when the cover **151** is open. The liquid drained from the intermediate box **700** may flow into the waste liquid box **60** as indicated by an arrow Y**6** in FIG. **9**. As shown in FIG. **11**A, the waste liquid box **60** has an opening **61**, a second waste liquid absorber **62**, and a second waste liquid reservoir **68**. The opening **61** is formed at a position to coincide with the protrusion **74** of the intermediate box **700**.
(119) The liquid in the intermediate box **700** may be passed to the second waste liquid reservoir **68** in the waste liquid box **60** through the protrusion **74**. In other words, the second waste liquid reservoir **68** may store the liquid passed from the waste liquid collecting devices. The waste liquid box **60** has a second waste liquid absorber **62** inside. The second waste liquid absorber **62** may absorb the liquid passed from the protrusion **74** of the intermediate box **700**. The second waste liquid absorber **62** may be made of, for example, unwoven fabric, sponge, or cotton that may absorb the liquid.
(120) As shown FIG. **11**A, the second waste liquid absorber **62** is in contact with the protrusion **74** of the intermediate box **700**. The second waste liquid absorber **62** has a contact surface **63**, on which the protrusion **74** abuts. With the second waste liquid absorber **62** having the contact surface **63**, the liquid absorbed in the first waste liquid absorber **73** in the intermediate box **700** may be passed from the protrusion **74** to the second waste liquid absorber **62**. Therefore, the liquid drained from the protrusion **74** may be stored in the second waste liquid reservoir **68**.
(121) According to the embodiment described above, the inkjet printer **200** may once collect the waste liquid produced in the inkjet printer **200** in the intermediate box **700** and thereafter store the waste liquid in the waste liquid box **60**.
(122) While the invention has been described in conjunction with example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of

the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents.

Claims

1. A liquid discharging apparatus, comprising: a main housing; a liquid discharging head having nozzles for discharging liquid; a container configured to store a plurality of sheets of printable medium inside the main housing; a waste liquid storage located at a position to at least partly overlap the container and lower than the container in a vertical direction, the vertical direction intersecting orthogonally with a nozzle surface of the liquid discharging head, the waste liquid storage being configured to store the liquid discharged from the liquid discharging head; and a mountable compartment, in which the waste liquid storage is demountably mountable, the mountable compartment being formed on a sideward face of the main housing on one side in a widthwise direction, the widthwise direction being a scanning direction of the liquid discharging head, the mountable compartment being open at a position frontward with respect to a connector, the connector connecting a power cord for receiving external power supply to the main housing there-through, in a front-rear direction being a conveying direction to convey the printable medium and toward which the printable medium is ejected from the main housing.
2. The liquid discharging apparatus according to claim 1, further comprising a liquid tank configured to store the liquid to be supplied to the liquid discharging head, wherein the liquid tank is located alongside the container and the waste liquid storage in the widthwise direction in the main housing.
3. The liquid discharging apparatus according to claim 1, further comprising a liquid tank configured to store the liquid to be supplied to the liquid discharging head, wherein the mountable compartment is located on the sideward face on a side in the widthwise direction opposite to the liquid tank across a conveyer path for the printable medium.
4. The liquid discharging apparatus according to claim 1, further comprising a liquid tank configured to store the liquid to be supplied to the liquid discharging head, wherein the waste liquid storage is located at least partly rearward in the front-rear direction with respect to a position to locate the liquid tank in the main housing.
5. The liquid discharging apparatus according to claim 1, wherein the mountable compartment is located on the sideward face between a first position, the first position being apart by a predetermined distance from a position where a rearward face of the main housing being on a rear side in the front-rear direction and the sideward face of the main housing meet, and a second position, the second position being apart by a predetermined distance from a position where a frontward face of the main housing being on a front side in the front-rear direction and the sideward face of the main housing meet.
6. The liquid discharging apparatus according to claim 1, wherein a dimension of the waste liquid storage in the widthwise direction is greater than a dimension of the container in the widthwise direction.
7. The liquid discharging apparatus according to claim 1, wherein the waste liquid storage is mounted in the mountable compartment in a posture, in which an edge of a frontward face of the waste liquid storage being on a front side in the front-rear direction toward an opening of the mountable compartment is located frontward with respect to an edge of the frontward face opposite to the opening of the mountable compartment.
8. The liquid discharging apparatus according to claim 1, wherein the mountable compartment is open at a position frontward with respect to a gripper for a user to grab the main housing.
9. The liquid discharging apparatus according to claim 1, further comprising a waste liquid

collecting device configured to collect the liquid discharged from the liquid discharging head.

10. The liquid discharging apparatus according to claim 9, wherein the waste liquid collecting device is a duct configured to suction mist of the liquid produced inside the main housing.

11. The liquid discharging apparatus according to claim 9, wherein the waste liquid collecting device is a platen configured to support the printable medium being conveyed.

12. The liquid discharging apparatus according to claim 9, wherein the waste liquid collecting device is a cap configured to seal the nozzles of the liquid discharging head by covering the nozzle surface.

13. The liquid discharging apparatus according to claim 9, wherein the waste liquid collecting device is a wiper configured to remove the liquid from the nozzles.

14. The liquid discharging apparatus according to claim 9, wherein the waste liquid collecting device is a receiver configured to receive the liquid discharged from the liquid discharging head in a flushing process.

15. The liquid discharging apparatus according to claim 9, further comprising an intermediate box having an outlet for ejecting the liquid collected by the waste liquid collecting device into the waste liquid storage.